

BHARAT INTERNATIONAL SENIOR SECONDARY SCHOOL

SCIENCE • Subject Code 086 CLASS X • UNIT TEST • 2026-27

Life Processes • Metals and Non-metals • Light: Reflection and Refraction

Date : _____ **Time Allowed : 3 hours** **Maximum Marks : 80**

General Instructions :

Read the following instructions carefully and follow them :

- (i) This question paper contains 39 questions. All questions are compulsory.
- (ii) The question paper is divided into three sections: A, B and C.
Section A : Biology (30 marks) Section B : Chemistry (25 marks) Section C : Physics (25 marks)
- (iii) The question paper has MCQs, VSAs, SAs, LAs and CBQs. Marks are given against each question.
- (iv) There are Case Based Questions (CBQs) with three sub-questions and are of 4 marks each.
- (v) Divide your answer sheet into three sections as per the question paper: Section A (Biology), Section B (Chemistry) and Section C (Physics). It is compulsory to answer each question in its respective section. Do not mix answers of one section into the other section.
- (vi) Instructions are given with each section and question, wherever necessary.
- (vii) Wherever a question requires a diagram, an alternative question has been provided for visually impaired candidates.
- (viii) There is no overall choice in the question paper. However, an internal choice has been provided in a few questions. Only one of the choices in such questions must be attempted.

SECTION A

BIOLOGY

1. A destarched potted plant bearing variegated leaves (partly green and partly white) was kept in sunlight for six hours. A leaf was then plucked, boiled in alcohol and tested with iodine solution. Which region of the leaf turned blue-black, and what does the result establish ? 1
(A) The green region ; chlorophyll is necessary for photosynthesis
(B) The white region ; chlorophyll is not necessary for photosynthesis
(C) Both the regions ; starch is transported across the leaf
(D) Neither region ; the leaf had already been destarched
2. An athlete develops cramps in the leg muscles after a sudden burst of running. Select the option that correctly shows the breakdown of glucose in the muscle cells under this condition. 1
(A) Glucose → Pyruvate → Ethanol + Carbon dioxide + Energy
(B) Glucose → Pyruvate → Carbon dioxide + Water + Energy
(C) Glucose → Pyruvate → Lactic acid + Energy
(D) Glucose → Lactic acid → Pyruvate + Energy
3. Which of the following correctly matches a secretion with its source and its action on food ? 1
(A) Trypsin - pancreas - emulsifies fats
(B) Bile juice - liver - digests proteins
(C) Salivary amylase - salivary glands - digests fats
(D) Pepsin - gastric glands - digests proteins
4. In which of the following animals is the mixing of oxygenated and deoxygenated blood most likely to take place, and why ? 1
(A) Fishes, because the heart has two chambers

- (B) Amphibians, because the heart has three chambers
 (C) Birds, because the heart has four chambers
 (D) Human beings, because the heart has four chambers
5. The table below shows the concentration (in %) of three substances measured at three different sites, X, Y and Z, associated with a human nephron. 1

Substance	Site X	Site Y	Site Z
Protein	8.00	0.00	0.00
Glucose	0.10	0.10	0.00
Urea	0.03	0.03	2.00

- Select the option that correctly identifies X, Y and Z.
- (A) X - filtrate in Bowman's capsule, Y - blood in glomerulus, Z - urine
 (B) X - blood in glomerulus, Y - urine, Z - filtrate in Bowman's capsule
 (C) X - urine, Y - filtrate in Bowman's capsule, Z - blood in glomerulus
 (D) X - blood in glomerulus, Y - filtrate in Bowman's capsule, Z - urine
6. A complete ring of bark was removed from around the stem of a healthy tree. For several days afterwards the leaves continued to receive water, but the roots gradually began to die. The best explanation for this observation is that : 1
- (A) the phloem lies in the bark, so the downward transport of food stopped
 (B) the xylem lies in the bark, so the upward movement of water stopped
 (C) transpiration pull is generated in the bark of the stem
 (D) the roots absorb food directly from the soil through the bark
7. The lungs of an adult human being contain millions of alveoli, giving a total surface area of about 80 m². The significance of this large surface area is that it : 1
- (A) increases the total volume of air that the lungs can hold
 (B) provides an extensive surface for the exchange of gases
 (C) warms the incoming air before it reaches the blood
 (D) prevents the collapse of the air passage during breathing
- For Questions number 8 and 9, two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.
- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.
8. Assertion (A) : The wall of the left ventricle is thicker than that of the right ventricle. 1
 Reason (R) : The left ventricle pumps blood to the lungs at a low pressure.
9. Assertion (A) : Lymph flows through the same vessels as blood and is pumped by the heart. 1
 Reason (R) : Lymph carries digested and absorbed fat from the intestine into the blood.
10. Why is the rate of breathing in aquatic organisms much faster than that in terrestrial organisms ? Name the respiratory pigment present in human blood and state the property that makes it suitable for its function. 2
11. The inner lining of the small intestine bears a very large number of finger-like projections. Name these projections and state two ways in which they help in the absorption of digested food. 2
12. Attempt either option (a) or (b) :

- (a) Glucose is present in the filtrate formed in Bowman's capsule but is absent in the urine of a healthy person. Explain why. Name any one factor on which the reabsorption of water from the filtrate depends. 2

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- (b) Plants do not possess any specialised excretory organ, yet they get rid of their waste products. State any two ways in which plants excrete their waste products. 2
13. During respiration, glucose is first broken down in the cytoplasm of the cell into a three-carbon molecule.
- (a) Name this three-carbon molecule and the process by which it is formed. 1
- (b) Name the organelle in which this molecule is broken down when oxygen is available, and write the end products formed in it. 1
- (c) Write the end products formed when this molecule is broken down in yeast in the absence of oxygen, and state how the energy released compares with that in part (b). 1
14. Name the tissue that transports the food prepared in the leaves to the other parts of a plant, and the process by which this transport takes place. State how this transport differs from the transport of water in a plant with respect to (i) the requirement of energy, and (ii) the direction of movement. 3

15. Read the following passage and answer the questions that follow :

A student took four test tubes P, Q, R and S. In each test tube she placed 1 mL of starch solution and 1 mL of freshly collected saliva. The four test tubes were kept for 15 minutes, each at a different temperature. A few drops of iodine solution were then added to each test tube and the colour was recorded.

Test tube	Temperature maintained	Colour with iodine solution
P	0 °C	Blue-black
Q	37 °C	No blue-black colour
R	60 °C	Blue-black
S	100 °C	Blue-black

- (a) In which test tube has starch been digested ? Name the enzyme responsible for this digestion. 1
- (b) Name the product formed from starch by the action of this enzyme. 1
- (c) (i) Explain why no digestion of starch has taken place in test tubes P, R and S. 2

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- (c) (ii) The contents of test tube Q were then made strongly acidic by adding dilute hydrochloric acid. State, with reason, the effect this would have on the action of the enzyme. Name the enzyme that acts on food under such acidic conditions and the food component it acts upon. 2
16. Attempt either option (a) or (b) :
- (a) (i) Draw a sectional view of the human heart and label the following parts : right atrium, left ventricle, pulmonary artery, aorta. 2
- (ii) Trace the path of a drop of deoxygenated blood, starting from the time it enters the heart from the body until it leaves the heart as oxygenated blood. 2
- (iii) Why are the walls of the ventricles thicker than the walls of the atria ? 1

O R

- (b) (i) Name the two conducting tissues present in a plant and state what each of them transports. 2
- (ii) Explain how the loss of water by transpiration helps in the upward movement of water in a tall plant. 2
- (iii) State one advantage, other than transport, that transpiration provides to a plant. 1

SECTION B
CHEMISTRY

17. An element X is so soft that it can be cut with a knife and is stored under kerosene. When a small piece of X is dropped into water, the hydrogen gas evolved catches fire immediately. The element X is most likely : 1

(A) Magnesium (B) Zinc
(C) Sodium (D) Iron

18. Four metals W, X, Y and Z were separately added to the salt solutions given below. A tick (✓) indicates that a reaction took place and a cross (X) indicates that no reaction took place. 1

Metal	ZnSO ₄ (aq)	FeSO ₄ (aq)	CuSO ₄ (aq)	AgNO ₃ (aq)
W	X	X	✓	✓
X	✓	✓	✓	✓
Y	X	X	X	X
Z	X	X	X	✓

The metals arranged in the decreasing order of their reactivity are :

(A) W > X > Y > Z (B) Y > Z > W > X
(C) X > Z > W > Y (D) X > W > Z > Y

19. Sodium chloride does not conduct electricity in the solid state, but conducts electricity in the molten state. This is because, in the molten state : 1

(A) the ions are converted into neutral atoms
(B) the ions become free to move about
(C) the ionic bonds are replaced by covalent bonds
(D) sodium releases electrons into the liquid

20. Which of the following pairs correctly matches an ore with the process used to convert it into its oxide ? 1

(A) Zinc carbonate - Calcination (B) Zinc carbonate - Roasting
(C) Zinc sulphide - Calcination (D) Mercury sulphide - Calcination

21. Which of the following metal oxides reacts with hydrochloric acid as well as with sodium hydroxide solution to produce salt and water ? 1

(A) Na₂O (B) CuO
(C) Al₂O₃ (D) MgO

22. An iron nail was kept for one week in each of the four test tubes described below. Rusting of the nail would be observed in : 1

(A) boiled distilled water covered with a layer of oil
(B) dry air over anhydrous calcium chloride
(C) a tube from which all the air has been removed
(D) tap water open to the air

23. The reaction $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow \text{Al}_2\text{O}_3(\text{s}) + 2\text{Fe}(\text{l}) + \text{Heat}$ is used to join broken railway tracks. In this reaction, aluminium acts as : 1

(A) an oxidising agent (B) a reducing agent
(C) a catalyst (D) an electrolyte

For Question number 24, two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.
24. Assertion (A) : Copper articles lose their shine and acquire a green coating when left exposed to moist air for a long time. 1
 Reason (R) : Copper reacts with moist carbon dioxide present in the air to form basic copper carbonate.
25. Write balanced chemical equations for the reactions that take place when : 1+1
 (i) zinc reacts with dilute hydrochloric acid.
 (ii) aluminium metal reacts with steam.
26. Attempt either option (a) or (b) :
 (a) A metal A is a shiny solid that is stored under kerosene. It cannot be obtained from its molten chloride by reduction with carbon.
 (i) Name the process by which A is obtained from its molten chloride. 1
 (ii) Write the reactions that take place at the cathode and at the anode during this process, taking A to be sodium. 2
- O R**
- (b) Give reasons for the following : 3
 (i) Ionic compounds have high melting points and boiling points.
 (ii) Aluminium is a highly reactive metal, and yet it is used to make cooking utensils.
 (iii) Silver articles turn black when kept exposed to air for a long time.
27. Four metals P, Q, R and S were tested and the following observations were recorded :
 (i) P reacts vigorously with cold water.
 (ii) Q does not react with cold water, but reacts with steam.
 (iii) R does not react with water at all, but reacts with dilute hydrochloric acid.
 (iv) S reacts neither with water nor with dilute hydrochloric acid.
 (a) Arrange P, Q, R and S in the decreasing order of their reactivity. 1
 (b) Name one metal each that Q and S could be. 1
 (c) Write the balanced chemical equation for the reaction of Q with steam. 1
28. Read the following passage and answer the questions that follow :
 A metal M of medium reactivity occurs in nature as its sulphide ore. The ore is first crushed and its gangue removed. The concentrated ore is then converted into the oxide of M, which is subsequently reduced to obtain the metal in an impure form. The impure metal so obtained is finally purified. The steps may be summarised as :
Sulphide ore $\rightarrow$ [Step 1 : Concentration] $\rightarrow$ *Concentrated ore* $\rightarrow$ [Step 2] $\rightarrow$ *Oxide of M* $\rightarrow$ [Step 3] $\rightarrow$ *Impure metal M* $\rightarrow$ [Step 4] $\rightarrow$ *Pure metal M*
 (a) Name the process carried out in Step 2 and write the balanced chemical equation for the conversion of zinc sulphide into zinc oxide by this process. 1
 (b) Name a suitable substance used to bring about Step 3 and write the balanced chemical equation for the reduction of zinc oxide by it. 1
 (c) (i) Name the process used in Step 4. State what is taken as the anode, as the cathode and as the electrolyte when zinc is purified by this process. 2

O R

- (c) (ii) When impure copper is purified by the process used in Step 4, state where the pure copper is deposited and where the impurities present in the impure copper collect. Give a reason for the second observation. 2

29. Attempt either option (a) or (b) :

- (a) (i) What is meant by corrosion ? Write the common name and the chemical formula of the green substance formed on copper articles exposed to moist air. 2
 (ii) State any two methods used to prevent the rusting of iron. 1
 (iii) What is an alloy ? Name the constituents of stainless steel and state one property of stainless steel which pure iron does not possess. 2

O R

- (b) (i) Draw the electron-dot structure to show the formation of magnesium chloride from magnesium (atomic number 12) and chlorine (atomic number 17). Name the type of bond formed in it. 3
 (ii) State any two properties of magnesium chloride that arise from this type of bonding. 2

SECTION C

PHYSICS

30. A concave mirror of focal length 15 cm forms an image which is real, inverted and of the same size as the object. The distance of the object from the mirror is : 1

- (A) 7.5 cm (B) 15 cm
 (C) 30 cm (D) 45 cm

31. Nikhil wrote down the following statements about a convex lens : 1

- (i) It always forms a virtual image, whatever the position of the object.
 (ii) It forms a real, inverted image of the same size as the object when the object is placed at $2F_1$.
 (iii) Its power is positive.
 (iv) Its focal length is taken as negative in the New Cartesian Sign Convention.

Choose the option enlisting all the correct statements :

- (A) (i) and (iv) (B) (ii) and (iii)
 (C) (i), (ii) and (iii) (D) (ii), (iii) and (iv)

For Question number 32, two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.

32. Assertion (A) : A ray of light travelling from water into glass bends towards the normal at the surface of separation. 1

Reason (R) : The refractive index of glass is greater than that of water, and light therefore slows down on entering the glass.

33. The refractive index of diamond is 2.42 and that of water is 1.33.

- (a) What is meant by the statement “the refractive index of diamond is 2.42” ? 1
 (b) In which of these two media does light travel faster ? Justify your answer. 1

34. Attempt either option (a) or (b) :

- (a) A concave lens has a focal length of 20 cm. At what distance from the lens should an object 5 cm tall be placed so that its image is formed 15 cm from the lens ? Also find the size of the image. 2

O R

- (b) A 2 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm. The distance of the object from the lens is 15 cm. Find the nature, position and size of the image formed. 2
35. (a) State the two laws of reflection of light. 2
- (b) A ray of light falls on a plane mirror in such a way that the angle between the incident ray and the reflected ray is 60° . Find the angle of incidence. 1
36. (a) While using the New Cartesian Sign Convention for a concave mirror, state the sign given to the object distance and the sign given to the focal length. 1
- (b) An object 4 cm high is placed at a distance of 30 cm in front of a concave mirror of focal length 20 cm. Find the position, the nature and the size of the image formed. 2
37. A convex mirror, and not a plane mirror, is used as the rear-view mirror in vehicles. State two reasons for this preference. Also state, with reason, why a convex mirror cannot be used by a dentist to examine the teeth of a patient. 3
38. Read the following passage and answer the questions that follow :

A student traced the path of a ray of light passing through a rectangular glass slab. She fixed the slab on a sheet of paper, drew its outline, and repeated the experiment for three different angles of incidence. Her readings are recorded below.

Angle of incidence $\angle i$	Angle of refraction $\angle r$	Angle of emergence $\angle e$
30°	19°	30°
45°	28°	45°
60°	35°	60°

- (a) What is the relationship between the angle of incidence and the angle of emergence in each case ? Give a reason for this relationship. 1
- (b) The emergent ray is shifted sideways with respect to the direction of the incident ray. State how this sideways shift changes if a thicker glass slab is used, the angle of incidence remaining the same. 1
- (c) (i) Using the readings recorded for $\angle i = 30^\circ$, calculate the refractive index of the glass of the slab. (Given : $\sin 30^\circ = 0.500$, $\sin 19^\circ = 0.326$) 2
- O R
- (c) (ii) The speed of light in vacuum is 3×10^8 m/s. Calculate the speed of light in a glass of refractive index 1.5. State whether light travels faster in this glass or in vacuum. 2
39. Attempt either option (a) or (b) :
- (a) (i) Draw ray diagrams to show the formation of the image by a convex lens when the object is placed (I) between F_1 and the optical centre O, and (II) beyond $2F_1$. 2
- (ii) An object is placed at a distance of 30 cm from a convex lens. A real and inverted image is formed at a distance of 60 cm from the lens on the other side. Calculate the focal length of the lens, the magnification produced and the power of the lens. 3

O R

- (b) (i) Define the principal focus of a concave mirror. Draw a ray diagram to show how a concave mirror converges a parallel beam of light incident on it. 2
- (ii) An object is placed at a distance of 10 cm in front of a convex mirror of radius of curvature 30 cm. Find the position, the nature and the magnification of the image formed. 3